

STRUTHIO SLIM4 · R22

Reverse-engineered schematic reference · generated from the R22 KiCad PCB connectivity, BOM and official component references

What this is: an auditable reconstruction of the board's pad-to-net connectivity, grouped into circuit areas. It captures the electrical connections stored in the native PCB layout. **It is not the original editable KiCad schematic** and does not restore the designer's original symbol graphics, annotations or design intent.

LCD validation flag: R22 J1 is an FH12 20-position, 0.5 mm connector. Startek's public KD047HDFID001 page lists a 31-pin, 0.6 mm interface. Treat this as an unresolved panel/FPC mismatch until the supplied panel drawing confirms the exact plug, pitch, pinout and flex orientation. The PCB-side J1 pin nets are recorded in the CSV.

Source counts: 166 placed components; 594 PCB pad records; 115 named/unnamed board nets. The CSV is the authoritative exhaustive connection listing.

R22 hardware signal summary

MCU	U1 · ESP32-P4NRW32X; logical nets include BTN_LEFT/RIGHT, DART_LEFT/RIGHT, I2S_BCLK/LRCLK/DOUT, BACKLIGHT_PWM, LCD_RESET_GATE, USB recovery, UART0, boot straps, flash, DSI and power rails.
Flash	U2 · W25Q512; FLASH_CS_N, FLASH_IO0..3, FLASH_CLK.
Display	J1 · 20 contacts; 2-lane DSI D0/D1/CLK, grounds, LCD reset/VCI/1V8, LED backlight pins.
Audio	U8/U9 · MAX98357A; shared I2S bus and separate AUDIO_SD_L/R, SPK_L+/-, SPK_R+/-.
USB	J2 USB-C; U13 TUSB320; U11/U12 USB ESD; USB VBUS/CC/DP/DM and recovery signals.
Controls	SW1..SW4 gameplay nets to ground; SW5..SW7 service controls.

Netlist format: one row per physical footprint pad. Blank net labels are marked "(no net)". The board-native connectivity is the source of truth; verify component pin functions and the display tail against manufacturer documentation before firmware or fabrication sign-off.

01 · P4, boot, flash and clock

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
C101 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	C126 C0805C106K8PAC7800	1→3V3_SYS; 2→GND
C102 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	C127 CL05B104KO5NNNC	1→3V3_SYS; 2→GND
C103 CL05B104KO5NNNC	1→1V1_HP; 2→GND	C128 CL05B104KO5NNNC	1→3V3_SYS; 2→GND
C104 CL05B104KO5NNNC	1→VDDO_FLASH_3V3; 2→GND	C129 CL05B104KO5NNNC	1→1V1_HP; 2→GND
C105 CL05B104KO5NNNC	1→1V1_HP; 2→GND	C130 CL05B104KO5NNNC	1→3V3_SYS; 2→GND
C106 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	C131 CL05B104KO5NNNC	1→3V3_SYS; 2→GND
C107 CL05B103KB5NNNC	1→VDDO_MIPi_2V5; 2→GND	C132 CL05B104KO5NNNC	1→3V3_SYS; 2→GND
C108 CL05B104KO5NNNC	1→VDDO_MIPi_2V5; 2→GND	C133 C0805C106K8PAC7800	1→3V3_SYS; 2→GND
C109 CL05A105KA5NQNC	1→VDDO_MIPi_2V5; 2→GND	C134 0402 C0G 22pF	1→1V1_HP; 2→FB_DCDC
C110 CL05B103KB5NNNC	1→VDD_USBPHY_LOCAL; 2→GND	C135 C0805X5R6R3-226MNE	1→1V1_HP; 2→GND
C111 CL05B104KO5NNNC	1→VDD_USBPHY_LOCAL; 2→GND	C136 CL05A105KA5NQNC	1→CHIP_PU; 2→GND
C112 0603X5R475K100NT	1→VDD_USBPHY_LOCAL; 2→GND	R103 RMCf0402FT4K02	1→DSI_REXT; 2→GND
C113 CL05B104KO5NNNC	1→VDDO_PSRAM_1V9; 2→GND	R104 RM04FTN4993	1→1V1_HP; 2→FB_DCDC
C114 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R105 RM04FTN4993	1→FB_DCDC; 2→GND
C115 CL05A105KA5NQNC	1→VDDO_FLASH_3V3; 2→GND	R106 0402WGF1002TCE	1→3V3_SYS; 2→CHIP_PU
C116 CL05B104KO5NNNC	1→VDDO_PSRAM_1V9; 2→GND	R107 0402WGF1002TCE	1→STRAP_GPIO34; 2→GND
C117 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R108 0402WGF1002TCE	1→3V3_SYS; 2→BOOT_STRAP
C118 CL05A105KA5NQNC	1→VDDO_PSRAM_1V9; 2→GND	R109 0402WGF1002TCE	1→3V3_SYS; 2→DOWNLOAD_STRAP_GPIO36
C119 CL05A105KA5NQNC	1→VDDO_PSRAM_1V9; 2→GND	R110 0402WGF1002TCE	1→3V3_SYS; 2→PWR_WAKE
C120 CL05A105KA5NQNC	1→VDDO_FLASH_3V3; 2→GND	R111 0402WGF1002TCE	1→3V3_SYS; 2→BTN_LEFT
C121 CL05A105KA5NQNC	1→VDDO_PSRAM_1V9; 2→GND	R112 0402WGF1002TCE	1→3V3_SYS; 2→BTN_RIGHT

Ref / Value	Pad → net	Ref / Value	Pad → net
C122 CL05A105KA5NQNC	1→VDDO_MIP1_2V5; 2→GND	U1 ESP32-P4NRW32X	1→BTN_LEFT; 2→BTN_RIGHT; 3→DART_LEFT; 4→DART_RIGHT; 5→I2S_BCLK; 6→I2S_LRCLK; 7→I2S_DOUT; 8→AUDIO_SD_CTRL; 9→3V3_SYS; 10→BACKLIGHT_PWM; 11→LCD_RESET_GATE; 12→CHG_STATUS; 13→NC; 14→BQ_EN1; 15→NC; 16→NC; 17→BAT_ADC; 18→USB_CURR_OUT2; 19→NC; 20→NC; 21→3V3_SYS; 22→NC; 23→NC; 24→NC; 25→NC; 26→1V1_HP; 27→FLASH_CS_SOC; 28→FLASH_IO1_SOC; 29→FLASH_IO2_SOC; 30→VDDO_FLASH_3V3; 31→FLASH_IO3_SOC; 32→FLASH_CLK_SOC; 33→FLASH_IO0_SOC; 34→DSI_REXT; 35→DSI_D1_P_SOC; 36→DSI_D1_N_SOC; 37→DSI_CLK_N_SOC; 38→DSI_CLK_P_SOC; 39→DSI_D0_P_SOC; 40→DSI_D0_N_SOC; 41→VDDO_MIP1_2V5; 42→NC; 43→NC; 44→NC; 45→NC; 46→NC; 47→NC; 48→NC; 49→NC; 50→NC; 51→VDD_USBPHY_LOCAL; 52→USB_JTAG_DM; 53→USB_JTAG_DP; 54→1V1_HP; 55→USB_RECOVERY_GPIO26; 56→USB_RECOVERY_GPIO27; 57→NC; 58→NC; 59→VDDO_PSRAM_1V9; 60→NC; 61→NC; 62→3V3_SYS; 63→NC; 64→NC; 65→STRAP_GPIO34; 66→BOOT_STRAP; 67→VDDO_PSRAM_1V9; 68→DOWNLOAD_STRAP_GPIO36; 69→UART0_TX; 70→UART0_RX; 71→VDDO_FLASH_3V3; 72→VDDO_PSRAM_1V9; 73→VDDO_MIP1_2V5; 74→NC; 75→3V3_SYS; 76→1V1_HP; 77→3V3_SYS; 78→FB_DCDC; 79→EN_DCDC; 80→NC; 81→NC; 82→NC; 83→NC; 84→USB_CURR_OUT1; 85→3V3_SYS; 86→PGOOD_STATUS; 87→NC; 88→BQ_EN2; 89→NC; 90→NC; 91→1V1_HP; 92→NC; 93→NC; 94→NC; 95→NC; 96→3V3_SYS; 97→NC; 98→NC; 99→XTAL_N_SOC; 100→XTAL_P_SOC; 101→3V3_SYS; 102→3V3_SYS; 103→CHIP_PU; 104→PWR_WAKE; 105→GND; 105→GND; 105→GND; 105→GND; 105→GND; 105→GND; 105→GND; 105→GND
C123 C0805C106K8PAC7800	1→3V3_SYS; 2→GND	U2 W25Q512JVEIQ TR	1→FLASH_CS_N; 2→FLASH_IO1; 3→FLASH_IO2; 4→GND; 5→FLASH_IO0; 6→FLASH_CLK; 7→FLASH_IO3; 8→VDDO_FLASH_3V3; EP→GND
C124 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	Y1 L327S400H11L	1→XTAL_P; 2→GND; 3→XTAL_N; 4→GND
C125 CL05B104KO5NNNC	1→1V1_HP; 2→GND		

02 · Power, charging and battery

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
C301 CL05B104KO5NNNC	1→SYS_RAW; 2→GND	R301 0402WGF0000TCE	1→DSI_CLK_P_SOC; 2→MIPI_DSI_CLK_P
C302 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R302 0402WGF0000TCE	1→DSI_CLK_N_SOC; 2→MIPI_DSI_CLK_N
C303 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R303 0402WGF0000TCE	1→DSI_D0_P_SOC; 2→MIPI_DSI_D0_P
C304 CL05B104KO5NNNC	1→LCD_1V8; 2→GND	R304 0402WGF0000TCE	1→DSI_D0_N_SOC; 2→MIPI_DSI_D0_N
C305 CL05A105KA5NQNC	1→3V3_SYS; 2→GND	R305 0402WGF0000TCE	1→DSI_D1_P_SOC; 2→MIPI_DSI_D1_P
C306 CL05A105KA5NQNC	1→LCD_1V8; 2→GND	R306 0402WGF0000TCE	1→DSI_D1_N_SOC; 2→MIPI_DSI_D1_N
C307 CL05A105KA5NQNC	1→3V3_SYS; 2→GND	R307 0402WGF1002TCE	1→LCD_1V8; 2→LCD_RESX
C308 CL05A105KA5NQNC	1→LCD_VCI_3V0; 2→GND	R308 0402WGF1003TCE	1→3V3_SYS; 2→LCD_RESET_GATE
C309 CL10A105KB8NNNC	1→LCD_LED_A; 2→GND	R309 0603WAF510KT5E	1→LCD_LED_K; 2→GND
C310 CL05B224KO5NNNC	1→BL_COMP; 2→GND	R401 0402WGF0000TCE	1→USB_JTAG_DP; 2→USB_DP_CONN
C311 0603X5R475K100NT	1→SYS_RAW; 2→GND	R402 0402WGF0000TCE	1→USB_JTAG_DM; 2→USB_DM_CONN
C401 CL05B104KO5NNNC	1→U4_VAUX; 2→GND	R403 0402WGF0000TCE	1→3V3_SYS; 2→VDD_USBPHY_LOCAL
C402 CL05B104KO5NNNC	1→SYS_RAW; 2→GND	R404 0402WGF0000TCE	1→CHASSIS_GND; 2→GND
C403 CL05B104KO5NNNC	1→USB_VBUS; 2→GND	R410 0402WGF4703TCE	1→3V3_SYS; 2→U4_FB
C404 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R411 0402WGF1503TCE	1→U4_FB; 2→GND
C405 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R412 0402WGF3601TCE	1→BQ_ISET; 2→GND
C406 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R413 0402WGF3301TCE	1→BQ_ITERM; 2→GND
C407 CL05B104KO5NNNC	1→3V3_SYS; 2→GND	R414 0402WGF4702TCE	1→BQ_TMR; 2→GND
C408 CL05A105KA5NQNC	1→3V3_SYS; 2→GND	R415 0402WGF1501TCE	1→BQ_ILIM; 2→GND
C409 0603X5R475K100NT	1→USB_VBUS; 2→GND	R416 0402WGF4703TCE	1→USB_VBUS; 2→USB_VBUS_SENSE_MID
C410 0603X5R475K100NT	1→3V3_SYS; 2→GND	R417 0402WGF4303TCE	1→USB_VBUS_SENSE_MID; 2→USB_VBUS_SENSE
C411 C0805C106K8PAC7800	1→SYS_RAW; 2→GND	R418 0402WGF1003TCE	1→BAT_PLUS; 2→BAT_ADC
C412 C0805C106K8PAC7800	1→BAT_PLUS; 2→GND	R419 0402WGF3302TCE	1→BAT_ADC; 2→GND
C413 C0805X5R6R3-226MNE	1→SYS_RAW; 2→GND	R420 0402WGF1003TCE	1→3V3_SYS; 2→CHG_STATUS
C414 C0805X5R6R3-226MNE	1→3V3_SYS; 2→GND	R421 0402WGF1003TCE	1→3V3_SYS; 2→PGOOD_STATUS
C415 C0805X5R6R3-226MNE	1→3V3_SYS; 2→GND	R422 0402WGF1003TCE	1→BACKLIGHT_PWM; 2→GND
C501 CL05B104KO5NNNC	1→SYS_RAW; 2→GND	R423 0402WGF1003TCE	1→SYS_RAW; 2→3V3_ENABLE
C502 CL05B104KO5NNNC	1→SYS_RAW; 2→GND	R424 0402WGF1002TCE	1→BQ_TS; 2→GND
C503 CL05B104KO5NNNC	1→SYS_RAW; 2→GND	R501 0402WGF2001TCE	1→AUDIO_SD_CTRL; 2→AUDIO_SD_L

Ref / Value	Pad → net	Ref / Value	Pad → net
C504 CL05B104KO5NNNC	1→SYS_RAW; 2→GND	R502 0402WGF1003TCE	1→AUDIO_SD_CTRL; 2→AUDIO_SD_R_MID
C505 CL05A105KA5NQNC	1→SYS_RAW; 2→GND	R503 0402WGF1103TCE	1→AUDIO_SD_R_MID; 2→AUDIO_SD_R
C506 CL05A105KA5NQNC	1→SYS_RAW; 2→GND	R504 0402WGF1003TCE	1→AUDIO_SD_CTRL; 2→GND
D2 STPS1L40ZFY	A→BL_SW; K→LCD_LED_A	U3 TLV62569DBVR	1→EN_DCDC; 2→GND; 3→CORE_SW; 4→3V3_SYS; 5→FB_DCDC
L1 MWSA0402S-1R0MT	1→U4_L1_SW; 2→U4_L2_SW	U4 TPS63070RNMR	1→SYS_RAW; 2→NC_U4_PG; 3→U4_VAUX; 4→GND; 5→U4_FB; 6→NC_U4_FB2; 7→3V3_SYS; 8→3V3_SYS; 9→U4_L2_SW; 10→GND; 11→U4_L1_SW; 12→SYS_RAW; 13→SYS_RAW; 14→3V3_ENABLE; 15→GND
L2 ASWPA4035S2R2MT	1→CORE_SW; 2→1V1_HP	U5 TLV75518PDBVR	1→3V3_SYS; 2→GND; 3→3V3_SYS; 4→NC; 5→LCD_1V8
L3 WPN4020H100MT	1→SYS_RAW; 2→BL_SW	U6 TLV75530PDBVR	1→3V3_SYS; 2→GND; 3→3V3_SYS; 4→NC; 5→LCD_VCI_3V0
Q1 2N7002	1→LCD_RESET_GATE; 2→GND; 3→LCD_RESX	U10 BQ24074RGTR	1→BQ_TS; 2→BAT_PLUS; 3→BAT_PLUS; 4→GND; 5→BQ_EN2; 6→BQ_EN1; 7→PGOOD_STATUS; 8→GND; 9→CHG_STATUS; 10→SYS_RAW; 11→SYS_RAW; 12→BQ_ILIM; 13→USB_VBUS; 14→BQ_TMR; 15→BQ_ITERM; 16→BQ_ISET; EP→GND

03 · Display connector and backlight

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
J1 FH12-20S-0.5SH(55)	1→LCD_RESX; 2→LCD_VCI_3V0; 3→LCD_1V8; 4→NC_PANEL_D3P; 5→NC_PANEL_D3N; 6→GND; 7→NC_PANEL_D2P; 8→NC_PANEL_D2N; 9→GND; 10→MIPI_DSI_CLK_P; 11→MIPI_DSI_CLK_N; 12→GND; 13→MIPI_DSI_D1_P; 14→MIPI_DSI_D1_N; 15→GND; 16→MIPI_DSI_D0_P; 17→MIPI_DSI_D0_N; 18→GND; 19→LCD_LED_K; 20→LCD_LED_A; MP→NC;	U7 TPS61165DBVR	1→SYS_RAW; 2→BACKLIGHT_PWM; 3→BL_SW; 4→GND; 5→BL_COMP; 6→LCD_LED_K

04 · Stereo audio

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
C201 CL05B104KO5NNNC	1→VDDO_FLASH_3V3; 2→GND	R205 0402WGF0000TCE	1→FLASH_IO2_SOC; 2→FLASH_IO2
C202 CL05A105KA5NQNC	1→VDDO_FLASH_3V3; 2→GND	R206 0402WGF0000TCE	1→FLASH_IO3_SOC; 2→FLASH_IO3
C203 0402CG120J500NT	1→XTAL_P; 2→GND	R207 0402WGF0000TCE	1→XTAL_P_SOC; 2→XTAL_P
C204 0402CG120J500NT	1→XTAL_N; 2→GND	R208 0402WGF0000TCE	1→XTAL_N_SOC; 2→XTAL_N
R201 0402WGF0000TCE	1→FLASH_CS_SOC; 2→FLASH_CS_N	R209 0402WGF1002TCE	1→VDDO_FLASH_3V3; 2→FLASH_CS_N
R202 0402WGF0000TCE	1→FLASH_CLK_SOC; 2→FLASH_CLK	U8 MAX98357AETE+T	1→I2S_DOUT; 2→GND; 3→GND; 4→AUDIO_SD_L; 5→NC; 6→NC; 7→SYS_RAW; 8→SYS_RAW; 9→SPK_L_P; 10→SPK_L_N; 11→GND; 12→NC; 13→NC; 14→I2S_LRCLK; 15→GND; 16→I2S_BCLK; EP→GND
R203 0402WGF0000TCE	1→FLASH_IO0_SOC; 2→FLASH_IO0	U9 MAX98357AETE+T	1→I2S_DOUT; 2→GND; 3→GND; 4→AUDIO_SD_R; 5→NC; 6→NC; 7→SYS_RAW; 8→SYS_RAW; 9→SPK_R_P; 10→SPK_R_N; 11→GND; 12→NC; 13→NC; 14→I2S_LRCLK; 15→GND; 16→I2S_BCLK; EP→GND
R204 0402WGF0000TCE	1→FLASH_IO1_SOC; 2→FLASH_IO1		

05 · USB-C and recovery

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
C601 CL05B104KO5NNNC	1→PWR_WAKE; 2→GND	R604 0402WGF1002TCE	1→BQ_EN2; 2→GND
C602 CL05B104KO5NNNC	1→RESET_MR; 2→GND	R605 0402WGF1002TCE	1→3V3_SYS; 2→USB_CURR_OUT1
C603 CL05B104KO5NNNC	1→BAT_ADC; 2→GND	R606 0402WGF1002TCE	1→3V3_SYS; 2→USB_CURR_OUT2
D1 PESD5V0S1UL,315	1→USB_VBUS; 2→GND	R607 0402WGF1002TCE	1→3V3_SYS; 2→RESET_MR
J2 USB4105-GF-A-120	→NC; →NC; A1→GND; A12→GND; A4→USB_VBUS; A5→USB_CC1; A6→USB_DP_CONN; A7→USB_DM_CONN; A8→NC_SBU1; A9→USB_VBUS; B1→GND; B12→GND; B4→USB_VBUS; B5→USB_CC2; B6→USB_DP_CONN; B7→USB_DM_CONN; B8→NC_SBU2; B9→USB_VBUS; SH→CHASSIS_GND; SH→CHASSIS_GND; SH→CHASSIS_GND	U11 TPD2EUSB30DRTR	1→USB_DP_CONN; 2→USB_DM_CONN; 3→GND
R601 0402WGF1002TCE	1→3V3_SYS; 2→DART_LEFT	U12 TPD2EUSB30DRTR	1→USB_CC1; 2→USB_CC2; 3→GND
R602 0402WGF1002TCE	1→3V3_SYS; 2→DART_RIGHT	U13 TUSB320LAIRWBR	1→USB_CC1; 2→USB_CC2; 3→GND; 4→USB_VBUS_SENSE; 5→NC_TUSB_ADDR; 6→NC_TUSB_OUT3; 7→USB_CURR_OUT1; 8→USB_CURR_OUT2; 9→NC_TUSB_ID; 10→GND; 11→GND; 12→3V3_SYS
R603 0402WGF1002TCE	1→3V3_SYS; 2→BQ_EN1		

06 · Gameplay and service controls

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
J3 SM02B-SRSS-TB(LF)(SN)	1→BAT_PLUS; 2→GND; MP→NC; MP→NC	SW3 D2LS-11	→NC; →NC; COM→DART_LEFT; NO→GND
J4 SM02B-SRSS-TB(LF)(SN)	1→SPK_L_P; 2→SPK_L_N; MP→NC; MP→NC	SW4 D2LS-11	→NC; →NC; COM→DART_RIGHT; NO→GND
J5 SM02B-SRSS-TB(LF)(SN)	1→SPK_R_P; 2→SPK_R_N; MP→NC; MP→NC	SW5 B3U-1000P	1→PWR_WAKE; 2→GND
SW1 D2LS-21(20M)	→NC; →NC; COM→BTN_LEFT; NO→GND	SW6 B3U-1000P	1→RESET_MR; 2→GND
SW2 D2LS-21(20M)	→NC; →NC; COM→BTN_RIGHT; NO→GND	SW7 B3U-1000P	1→BOOT_STRAP; 2→GND

07 · Other passives and connectors

Each row shows the pad-level net assignments recovered from the PCB. Repeated net names are electrically common in the board layout.

Ref / Value	Pad → net	Ref / Value	Pad → net
U14 TPS3808G33DBVR	1→CHIP_PU; 2→GND; 3→RESET_MR; 4→NC_U14_CT; 5→3V3_SYS; 6→3V3_SYS		